

Supplemental information

**Targeted expansion of cytotoxic T cells
using IL-12 and CD137L supplementation
enhances antitumor efficacy**

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Table S1. Characteristics of the healthy donor volunteers used throughout the study

Donor	Specimen	Sex	Age	Basal V δ 2
1	Healthy	Female	56	0.51
2	Healthy	Male	66	1.17
3	Healthy	Female	49	2.08
4	Healthy	n/a	n/a	2.41
5	Healthy	n/a	n/a	1.57
6	Healthy	Male	45	0.38
7	Healthy	n/a	n/a	1.28
8	Healthy	n/a	n/a	4.62
9	Healthy	Female	44	0.24
10	Healthy	Male	47	0.69
11	Healthy	Female	68	2.46
12	Healthy	Female	59	1.72
13	Healthy	Male	38	2.23
14	Healthy	Female	30	1.35
15	Healthy	Male	60	0.79
16	Healthy	Male	37	0.5
17	Healthy	Female	56	4.51
18	Healthy	Male	31	0.31
19	Healthy	Male	43	2.33
20	Healthy	Female	63	1.42
21	Healthy	Male	33	3.62
22	Healthy	Male	16	12.4
23	Healthy	Female	17	0.8
24	Healthy	Female	17	0.79
25	Healthy	Male	51	0.37
26	Healthy	Male	48	0.047
27	Healthy	Male	41	3.6
28	Healthy	Male	63	1.34
29	Healthy	Female	54	3.75
30	Healthy	Male	38	3.56
31	Healthy	Male	18	0.28
32	Healthy	Female	24	0.16
33	Healthy	Female	18	0.55
34	Healthy	Female	69	0.75
35	Healthy	Male	58	0.59
36	Healthy	Male	41	1.64
37	Healthy	Female	56	0.51
38	Healthy	Male	16	6.61
Mean			43.24	1.95

Participants 20-38 were included in cytotoxicity assays.

Table S2. Median of frequencies upon expansion with N-BPs and IL-2 for 10 days

	Concentration	0	0.08	0.15	0.3	0.6	1.5	2	3	4.5
ALN	Vδ2	1.4	9.4	23.1	31.5	54.9	67.3	64.2	70.3	47.0
	Vδ2 CD16	10.6	24.8	29.6	27.8	26.3	26.7	24.7	28.4	25.6
	Vδ2 CD56	6.4	41.3	37.7	40.7	37.0	30.3	29.2	23.2	23.2
	Vδ2 CD8	11.9	12.2	9.8	9.6	8.0	6.9	5.7	5.8	3.7
	Concentration	0	1.8	3.6	5.3	7	9	18	36	90
PAM	Vδ2	1.4	25.6	31.0	32.9	32.3	33.0	29.2	18.9	3.1
	Vδ2 CD16	10.6	20.9	16.9	20.0	21.0	22.3	28.3	28.4	33.1
	Vδ2 CD56	6.4	28.4	30.8	25.0	25.3	31.8	25.4	23.4	32.6
	Vδ2 CD8	11.9	7.0	6.3	6.0	6.4	5.6	5.6	7.4	10.1
	Concentration	0	0.5	1	1.5	2.0	2.5	5.0	10	25
ZOL	Vδ2	1.4	44.0	51.1	52.6	53.7	54.9	48.7	33.5	11.2
	Vδ2 CD16	10.6	30.2	17.7	27.0	35.2	23.1	28.9	26.2	24.6
	Vδ2 CD56	6.4	23.1	22.5	24.0	23.6	25.5	25.8	22.8	18.9
	Vδ2 CD8	11.9	6.2	5.9	5.2	6.1	5.1	6.8	7.1	7.3

Numbers indicate frequency (%) of CD3+ T cells. N=8

Table S3. Means of the frequencies and mean fluorescence intensity of V δ 2 T cells cultured with different conditions

Experiment	Conditions	V δ 2	V δ 2 CD16	V δ 2 CD56	V δ 2 NKG2A	V δ 2 NKG2D	V δ 2 GzmB	V δ 2 IFN- γ	Viability	V δ 2 CD16 GzmB	V δ 2 NKG2A GzmB
1	AIL2	86.86	28.63	30.01	45.30	8011.43	10501.71	2301.00	95.76	10914.71	11140.71
2	AIL2vitC	90.34	31.69	32.87	40.86	7309.57	7595.14	1350.29	97.26	7582.14	7350.71
3	AIL2CD277	91.21	24.46	26.46	35.96	7380.29	8408.86	1423.86	96.49	8169.00	8299.00
4	AIL2CD6	91.80	27.29	30.61	45.09	6214.14	9055.86	2672.86	93.89	9690.00	9578.29
5	AIL2CD137	91.40	28.73	31.36	48.50	6139.43	10576.86	3359.71	90.49	10664.57	10793.57
6	AIL212	78.84	25.61	34.97	49.77	5767.29	14300.43	2016.43	92.06	14927.14	14958.43
7	AIL212vitC	84.23	28.10	42.17	51.27	5574.14	10685.00	1486.71	94.40	10945.71	10848.29
8	AIL212CD277	82.44	21.28	30.70	43.16	5927.14	14321.00	1827.57	90.20	12509.57	14088.57
9	AIL212CD6	84.01	22.55	35.83	51.13	4704.00	13003.86	2806.00	88.63	13271.29	13695.00
10	AIL212CD137	81.17	25.01	36.01	51.51	4674.00	15164.29	3499.86	90.09	18343.71	19708.86
11	AIL221	82.47	25.61	29.89	44.76	6444.00	12427.86	2136.57	91.50	15821.00	15221.43
12	AIL221vitC	87.30	30.17	30.29	47.36	6510.29	10658.00	1465.14	93.44	10889.00	10934.14
13	AIL221CD277	85.57	20.83	25.13	41.69	6823.71	11249.57	1643.71	91.70	10689.86	11472.00
14	AIL221CD6	86.74	25.15	30.26	45.44	4930.57	12337.29	2989.00	88.94	12744.14	12766.29
15	AIL221CD137	86.37	26.02	31.47	46.17	5155.57	12791.29	2948.00	91.16	13473.71	13594.00
	MEAN	86.05	26.08	31.87	45.86	6104.37	11538.47	2261.78	92.40	12042.37	12296.62

Numbers indicate frequency (%) of CD3+ T cells for V δ 2, CD16, CD56, NKG2A and viability, and mean fluorescence intensity (MFI) for NKG2D, GzmB and IFN- γ . N=7

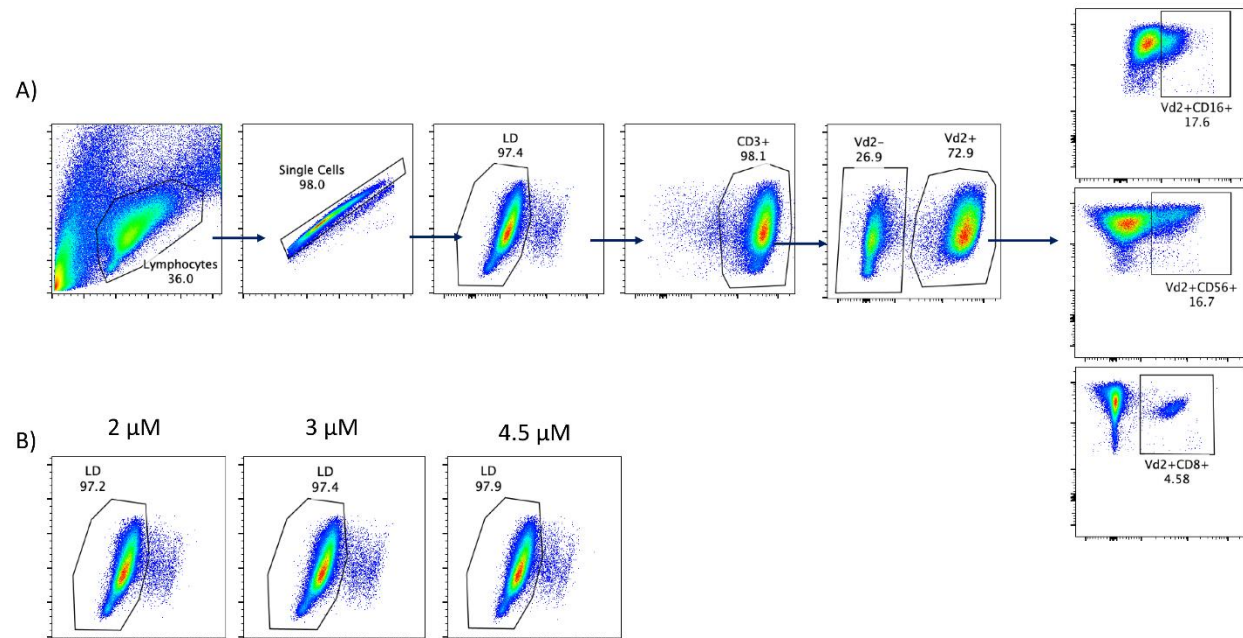


Figure S1: Representative example of the gating strategy

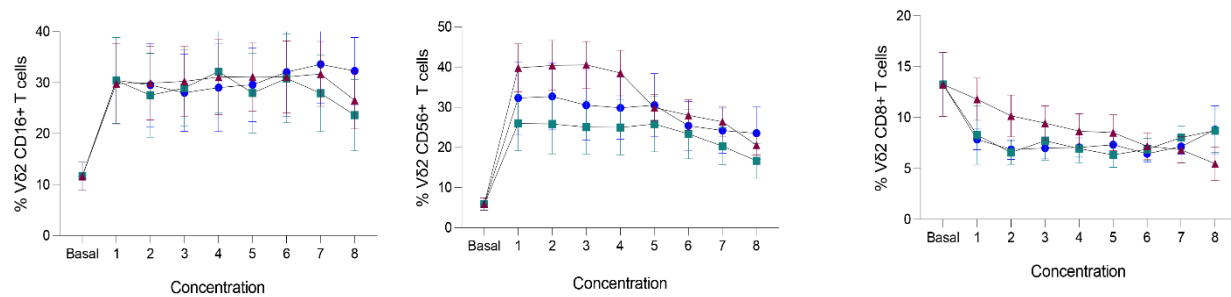


Figure S2: Frequency of Vδ2 T cells expressing CD16 (left), CD56 (middle) or CD8 (right) upon exposure to PAM (blue), ZOL (tile) or ALN (maroon) in cultures of 10 days. N=8.

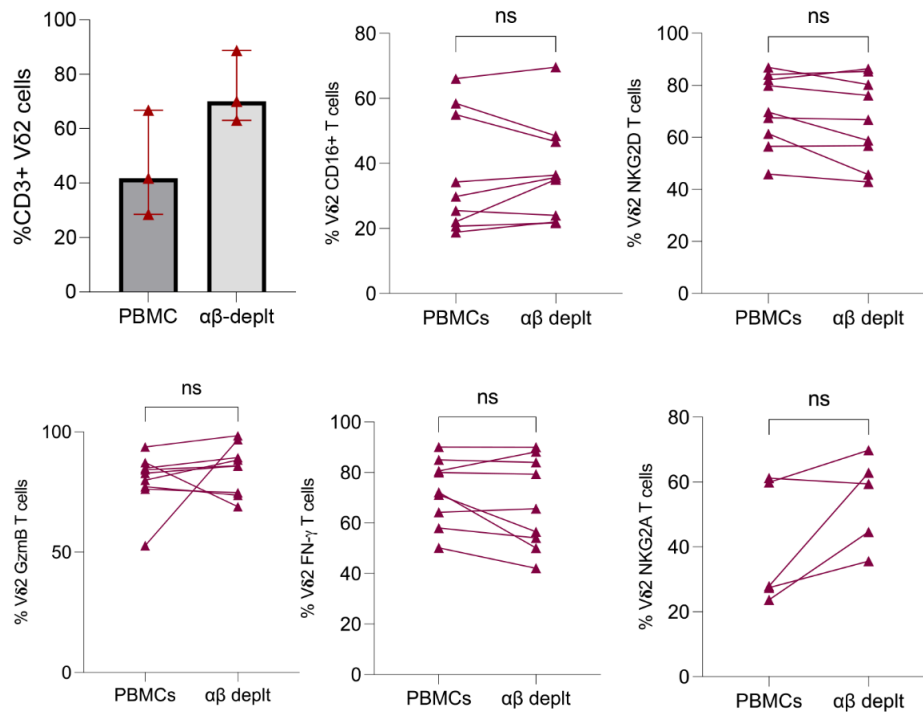


Figure S3: Frequency of total V δ 2 T cells and expression of cytotoxic markers on ALN-expanded V δ 2 T cells after 10 days in cultures that started from whole PBMC compared to $\alpha\beta$ -TCR+ depleted PBMCs.

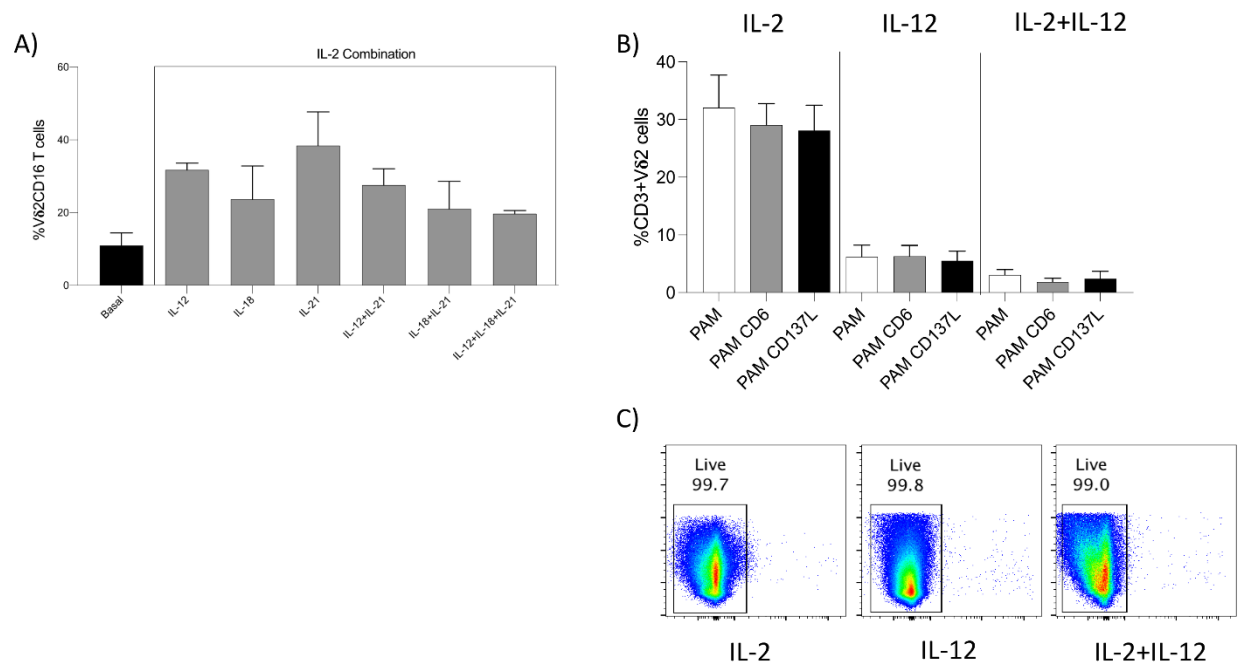


Figure S4: Expansion of VδT cells A) when IL-2 was combined with IL-12, IL-18 and IL-21. B) in the continuous presence of cytokines and UCD6 or CD137L. C) Viability of expanded Vδ2 T cells. N=4

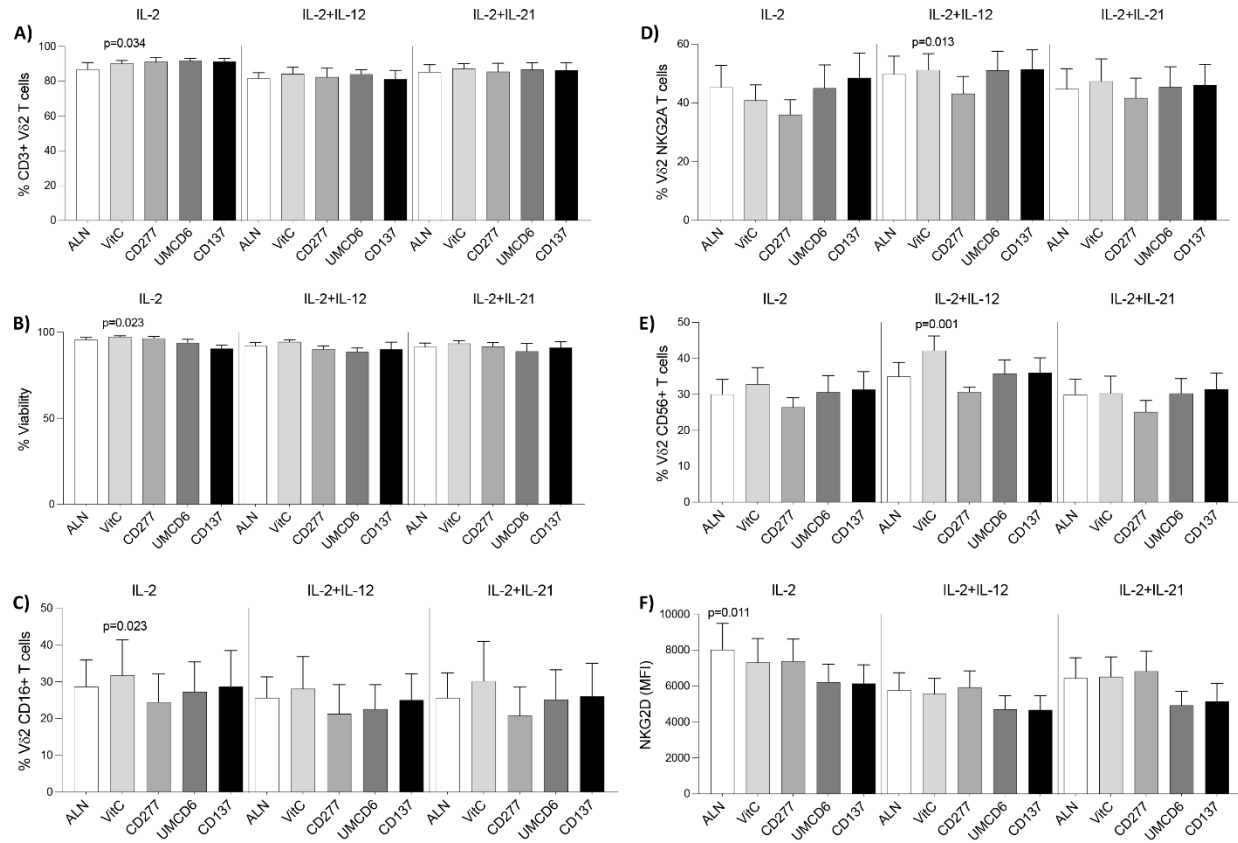


Figure S5: Comparison of Vδ2 T cell expansion after exposure to ALN+IL-2, ALN+IL-2+IL-12 or ALN+IL-2+IL-21 combined with either vitamin C, CD277, UCD6 or CD137L. A) Vδ2 frequency, B) viability, and expression of C) CD16, D) NKG2A, E) CD56 and F) NKG2D, in expanded Vδ2 T cells. N=7.

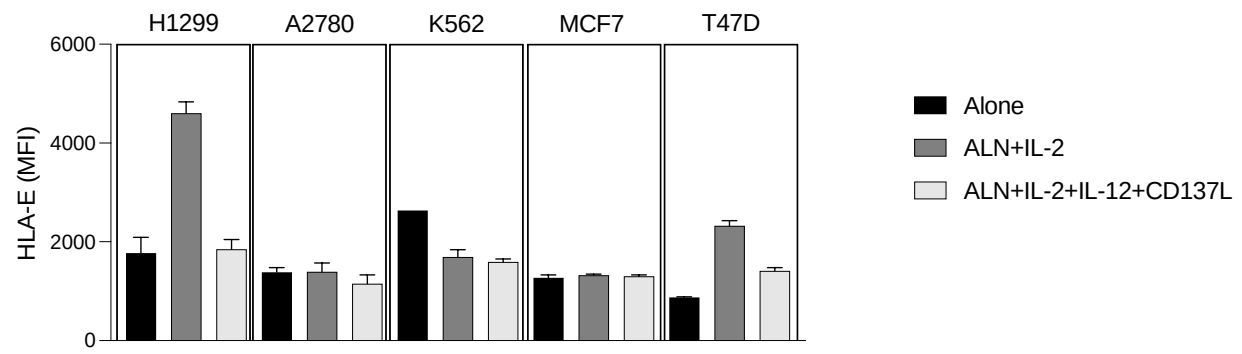


Figure S6: HLA-E expression in the different cancer cell lines